Australian capital gains tax research pack

Research snapshot: **11 November 2026**. Independent of the New Zealand packs. This is research and reproducible illustration, not an op-ed, tax-return calculator or predetermined argument.

- Open the 62-chart gallery
- Ten-year worked comparison: PV taxes and METRs, with 12 additional plots and a separate reproducible script
- Research guide and suggested reading paths
- Enacted rules, announcements and implementation work
- Behavioural evidence, definitions and original links
- Numbers for writing without charts
- Source notes and reporting breaks

Every chart has PNG, SVG and its exact plotted CSV. The searchable local gallery includes notes, source names and links to downloaded originals. Nothing is published externally.

Run

From `opeds`, using the installed R:

```
& "C:\Program Files\R\R-4.5.2\bin\Rscript.exe" "Australia_CGT\00_run_all.R" --cached
```

Options:

- No flag: download a fresh copy of each **pinned release URL**, then rebuild. This does not discover new annual releases or update the policy-status research automatically.
- `--cached`: use saved public-source files without network requests; rebuild source tables, models and plots.
- `--missing`: download only sources missing locally, then rebuild.
- `--plots-only`: reuse extracted source tables; rebuild models and plots.

`01_pull_data.R` handles public-source acquisition and extraction; `02_models.R` holds scenario assumptions and calculations; `03_make_plots.R` exports charts and the gallery. The runner stops on required-source or validation failures.

R dependencies: `dplyr`, `tidyr`, `readr`, `purrr`, `ggplot2`, `scales`, `svglite`. Python dependencies: `requests`, `beautifulsoup4`, `pandas`, `openpyxl`, `PyMuPDF`, and, for slide context extraction, `python-pptx` and `Pillow`. These were already available here; the scripts install nothing. Set `AU_CGT_PYTHON` to another interpreter if needed. The local Windows interpreter is detected as a convenience.

The Python helpers perform TLS-verified downloads, extract workbook/PDF tables and read slide text/pictures; **all thought-experiment calculations and plots are in R**. If requests cannot see a Windows enterprise root certificate, urllib's default verified Windows SSL context is used. Certificate verification is never disabled.

Two optional original papers (OBR and Dowd/McClelland) deny direct programmatic downloads. Their parameters were checked through the web research tool and transcribed with URLs, locations, versions and definitions in `literature.csv`. They are not represented as downloaded PDFs. `source_availability.csv` records this limitation, including during cached runs.

`parameters.csv` and `personal_tax_schedule.csv` are **audit outputs**, not configuration files. Change assumptions in `02_models.R` and the corresponding captions in `03_make_plots.R`, then rerun. Keep source/model labels and units when reusing a CSV.

Provenance and checks

- `processed/source_manifest.csv`: 32 downloaded public sources, URLs, sizes, timestamps and SHA-256 hashes.
- Source extracts retain worksheet/page and row/column locations where relevant.
- `processed/asset_reconciliation.csv`: differences between published ATO totals and their reported asset components; no forced rescaling.
- `processed/validation.csv`: source anchors, disjoint-group checks and accounting/model identities.
- `processed/output_validation.csv`: image, data, link and source-hash checks, written by `helpers/validate_outputs.py`.
- `figures/chart_catalogue.csv`: title, category, status, interpretation caveat and source for each chart.

The original slides remain unchanged. Their extracted text and contact sheets are in `processed/`; they are contextual research material and are not executed. Raw PDFs and slide material are provided for local research; link to the original publisher when sharing a blog post.

The source snapshot is dated; legal implementation details and the parliamentary position need rechecking before an article goes to press. Windows R may emit existing locale/package-build warnings; source scripts use ASCII with Unicode escapes to avoid corrupting labels.